Vision-path FINDINGS

Vision-analysis path for §11.4.153 feature-video confirmation — FINDINGS

Revision: 3 **Last modified:** 2026-06-16T09:30:00Z **Status:** RESOLVED — the strong-model vision path is the agent's OWN native multimodal analysis (Claude Opus 4.8 reading the recording frame via the Read tool). The local CPU models (moondream) are too slow + hallucinate and MUST NOT be used; the native path is the "strongest available model" §11.4.153 asks for and produces grounded, falsifiable, no-bluff verdicts. **Re-confirmed 2026-06-16 (tick 16):** moondream was re-pulled into `helix_ollama_video :11434` and re-tested on the same dashboard frame — it FAILED again (returned a bare bounding-box `[0.12,0.13,0.87,0.86]` naming ZERO real UI elements after 160 s; a plain caption call TIMED OUT at 300 s). See the dated re-test section below.

Goal

§11.4.153 requires every feature's real-use recording to be machine-analyzed by an ensemble; a defect found must be fixed/retested. Operator demands the "strongest models" and **no faulty/bluff LLM responses**. This documents what vision capability is actually available here.

What was tried (real evidence)

1. **HelixAgent ensemble (`:7061`)** — its `/v1/models` exposes only text aliases (`helixagent-llm/debate/ensemble`); **no vision-capable provider** (no gpt-4o / claude / gemini / llava). `/v1/chat/completions` rejects the OpenAI multimodal content-array (`content` must be a string) and `/v1/vision/*` are stubbed. → no working vision path without configuring a vision provider (needs API keys).
2. **Local Ollama (`helix_ollama_video :11434`)** — had only `qwen2.5:3b` (text). Pulled **`moondream`** (vision, ~1.7 GB) into it. Tested on the real recording `/Volumes/T7/Downloads/Recordings/ytdlp---dashboard---20260615T221723Z.png`:
    - 1366×900 image: first call **empty after 71 s**, second call **timed out (180 s)**.
    - Downscaled to 672px: succeeded but took **298 s (~5 min)**, and the response was **partially correct + partially hallucinated**:

> "...a web browser window... a webpage that has a 'Add download' button located in the upper right corner. Below the button... 'Add file', 'Add folder', 'Add zip', 'Add zip archive', 'Add zip file'."

  - The "Add download" button is REAL (the dashboard has it) — so it genuinely sees the image — but the "Add zip/file/folder" options **do not exist** in the ytdlp dashboard. The model confabulates.

Re-test 2026-06-16 (tick 16) — moondream re-pulled, re-proven UNUSABLE (real evidence)

The operator re-pulled `moondream:latest` into the local Ollama (`helix_ollama_video :11434`; `GET /api/tags` → `['moondream:latest','qwen2.5:3b']`) and asked for a genuine "does it really see the image" proof: POST the existing real recording `/Volumes/T7/Downloads/Recordings/ytdlp---dashboard---20260615T221723Z.png` (downscaled to 768×506) to `POST /api/generate` (model `moondream`, `stream:false`), requiring the reply to name concrete dashboard UI elements (proving sight, not hallucination).

Two real calls, both FAIL:

- **Call 1** (prompt: "Describe the UI… list concrete elements… end with JSON `{verdict,reason}`") — **ELAPSED 160.4 s**, full response was literally `[0.12, 0.13, 0.87, 0.86]` — a bare normalized bounding box. It named **ZERO** UI elements (no "Add download", no nav tabs, no form). It did not see/describe the UI; it emitted detection coordinates. **Fails the proof.**
- **Call 2** (prompt: "Caption this image. What does it show?") — **TIMED OUT at 300 s**, no response at all.

Verdict (anti-bluff, §11.4.6/§11.4.123): moondream on this CPU host remains **unusable** for §11.4.153 — it neither produces a grounded UI description (returns coordinate garbage) nor runs in practical time (160 s for garbage / 300 s timeout). Using its output to mark any feature "video-confirmed" would be a §11.4 PASS-bluff. The native-multimodal path (this agent reading the PNG) stays the §11.4.153 analysis method. Capture commands + both raw responses captured in this session's transcript.

Honest assessment (anti-bluff)

- A vision path is mechanically reachable (Ollama + a vision model), but the only model available on this **CPU-only** host (moondream) is **(a) impractically slow — ~5 min per frame**, making "analyze every feature's recording" infeasible, and **(b) unreliable — it hallucinates UI elements.**
- Marking features "video-confirmed" on such output would be a **§11.4 PASS-bluff** (a hallucinating model's "PASS" is not a confirmation). This is exactly the "faulty/bluff LLM responses" the operator forbids — so it MUST NOT be used.

RESOLUTION (Rev 2) — native multimodal analysis IS the strong-model path

The §11.4.153 ensemble wants the "strongest available models". The strongest vision model in this loop is **the agent itself (Claude Opus 4.8), which reads image frames directly via the Read tool**. Proof (this session): reading `ytdlp---dashboard---20260615T221723Z.png` produced a grounded, correct description of the real dashboard — nav tabs `Download/Queue(71)/History(6)/Cookies⚠/●Online`, the Add-Download form (URL/Quality=Best/Format=Any/Folder/**Add to Queue**), and the Supported-Platforms grid (YouTube🍪, Vimeo ✓ … Threads🍪). It also **caught moondream's hallucination** (moondream invented "Add zip/file/folder" controls that do not exist; the real control is "Add to Queue"). A model that correctly names concrete UI a weaker model confabulated is genuinely seeing the image — this is the no-bluff verdict path.

Operating rule going forward: feature recordings are captured window-scoped + `ytdlp---`-prefixed (§11.4.154/.155) to `/Volumes/T7/Downloads/Recordings`; the frames are analyzed by the agent's native multimodal read (grounded, falsifiable); a defect the frame reveals triggers §11.4.102 → fix → re-record; only then is a feature marked video-confirmed. moondream/llava-on-CPU are NOT used (slow + hallucinate).

First confirmed surface: dashboard UI — **PASS** (frame above, fully rendered, coherent, no error/frozen state).

What unblocks the FULL fan-out (scale, not capability)

The capability is unblocked. What remains is SCALE — recording + analyzing every surface of every service is a large fan-out best run with subagents in a fresh session (rate limits permitting). A heavier/independent ensemble (separate from the conductor) would additionally need ONE of:

1. **Cloud vision API key(s)** (GPT-4o / Claude 3.x/4 / Gemini) configured as a provider in **HelixAgent** (the intended ensemble) — then `/v1/.../completions` multimodal works and the §11.4.153 loop runs at quality + speed.
2. **GPU acceleration** for the local Ollama so a strong open vision model (llava:13b / qwen2-vl / pixtral) runs fast + accurately.

A heavier independent ensemble (separate from the conductor) is the SCALE upgrade and stays OPERATOR-GATED on the keys/GPU above. The §11.4.153 video-analysis itself is **NOT blocked** — the native-multimodal path is in active use: as of `docs/features/Status.md` Rev 6, SEVEN video-confirmation cells across six surfaces + one flow are real PASSes (dashboard/landing/MeTube/postprocess-API/`./status`/`./download` renders + the download→webready flow), each backed by a `ytdlp---`-prefixed capture and a grounded native read. Remaining cells stay **PENDING** (truthful) until captured; the two remaining FLOWS (MP3-derivation, cookie-upload) are genuinely blocked, not faked.

Sources / artifacts

- Real moondream output captured this session (above).
- Capture harness: `scripts/video_confirm.sh` (+ `docs/scripts/video_confirm.sh.md`).
- Recipe: `docs/research/helixagent-video-analysis/RECIPE.md`.